Takeme2space – Deep Dive

TakeMe2Space (TM2S) is a Hyderabad-based **SpaceTech/DeepTech company founded in 2023**, building orbital computing infrastructure designed to shift data processing closer to where satellite data is generated. Its central proposition is that satellite operators should not need to transmit large volumes of raw Earth-observation data to ground stations before extracting useful information; instead, AI models can execute onboard the satellite, with only the resulting intelligence transmitted to Earth. The company is therefore positioning itself not merely as another satellite-data provider, but as an **infrastructure layer for in-orbit compute and sensing.**

1. Founder & Team

Ronak Kumar Samantray is the founder and CEO. TakeMe2Space's current company biography states that he previously co-founded NowFloats, which was acquired by Reliance, and subsequently founded TakeMe2Space in 2023. Chiratae describes him as a seasoned entrepreneur with approximately 15 years of experience across software and frontier technology.

The broader team is unusually relevant to the physical-infrastructure problem. Anand Rajagopalan, previously CEO of T-Works, is associated with business strategy; Sharat Kamidi has a product/manufacturing background; and Firoz Ahammad has rapid-prototyping and engineering experience. Chiratae states that the team has built more than 15 satellite modules in-house and successfully conducted earlier missions to space-qualify technology.

The founder-market-fit thesis is therefore less about conventional aerospace pedigree and more about the combination of software thinking, product accessibility and rapid hardware iteration. The team is attempting to make space infrastructure behave more like a developer platform: customers interact with a service rather than owning the underlying spacecraft.

2. What Does TakeMe2Space Do - Their products.

The core problem is a data-location mismatch. Satellites can capture large quantities of Earth-observation data in orbit, but conventional architectures transmit raw data to ground infrastructure before analytics are performed. Downlink windows, ground-station availability, transmission costs and terrestrial processing create latency and bandwidth constraints.

TM2S moves part of the computation to the point where data is generated. A customer can upload an AI model, define an area of interest and have inference performed on the satellite. Instead of transmitting a full image set, the system can transmit a compact result - for example, detected objects, vegetation anomalies, changes or other structured outputs.

Products & Services

1. OrbitLab — Orbital Compute-as-a-Service

- TM2S's flagship platform for accessing **satellite compute and Earth-observation infrastructure without owning a satellite**.
- Customers can upload and execute AI models onboard the spacecraft, with processed intelligence transmitted back instead of large volumes of raw data.
- Current spacecraft specification: **117 TOPS onboard compute**, **9-band multispectral imaging**, and **~9 m ground resolution across a 19.15 km swath**.
- Company claims processed insights can reach users within **5–15 minutes of capture**.
- **Pricing:** $4/minute for live satellite utilisation; **$200/orbit for academic use**; sandbox access is free; enterprise volume/dedicated-capacity pricing is customised.

- Key applications: **change detection, object detection, anomaly detection, agriculture, mining, insurance and supply-chain intelligence**.
- **Business model:** usage-based recurring revenue, with potential to scale as satellite capacity and customer utilisation increase.

2. OrbitVault — Orbital Data Storage

- A proposed **space-based archival and cold-storage infrastructure** designed to provide physical separation from terrestrial data centres.
- Each satellite is stated to provide **200GB+ of NVMe storage**.
- Security architecture includes **encryption, per-tenant isolation, SHA-256 verification, CRC32 integrity checks and Reed-Solomon error correction**.
- Target applications include **financial archives, defence/government data, digital assets and sensitive corporate/M&A information**.
- **Business model:** enterprise-oriented storage service; public pricing is not currently disclosed.

3. Satellites & Subsystems — Space Hardware

- TM2S operates a vertically integrated hardware portfolio covering **36 products**, including complete satellite platforms and individual subsystems.
- Platforms range from **1U/2U CubeSats to 3U, 6U, 8U, 12U, 16U and 30–50 kg MicroSat platforms**.
- Public platform pricing ranges approximately from **₹25 lakh to ₹3 crore**, depending on platform size and configuration.
- Individual products include **AI accelerators, onboard computers, ADCS, reaction wheels, star trackers, power systems, communications modules, imaging components and radiation shielding**.
- **Business model:** primarily one-time hardware sales, with potential strategic value through vertical integration and space-qualification of TM2S's own technology.

4. Hosted Payload Missions — Space-as-a-Service

- Enables customers to fly **experiments, sensors and other payloads on TM2S spacecraft** without developing and launching an entire satellite.
- Current roadmap includes:
  - **MOI-1B:** 30 kg spacecraft; **6 kg payload capacity**; 20 W nominal / 50 W peak power; Q4 2026 target.
  - **AADUSat:** 100 kg; **28 kg payload capacity**; 200 W nominal / 700 W peak power; Q1 2027 target.
  - **MOI-4 & MOI-5:** 100 kg each; **15 kg payload capacity**; 200 W nominal / 700 W peak power; Q3 2028 target.
- Target customers include **research institutions, universities, startups and organisations seeking flight heritage or orbital testing**.
- **Business model:** payload hosting and mission-access fees; public pricing is not disclosed.

5. Manha — Space Education Platform

- A **1U CubeSat-standard educational satellite platform** designed for hands-on learning rather than simulation alone. Designed for **schools, universities, STEM programmes and individual learners**.
- Offers **20+ experiments, 3 programming languages and 4 real satellite subsystems**.
- Includes a satellite, ground-station receiver, antennas and software dashboard for monitoring satellite parameters.
- **Business model:** hardware/education product sales; public standard pricing is not disclosed.

Portfolio logic: The products collectively create multiple monetisation layers — **hardware sales → hosted payloads → recurring orbital compute → orbital storage**, with Manha providing an education/STEM customer channel. The key investment question is whether TM2S can shift its revenue mix over time from relatively episodic hardware and mission revenue toward **higher-margin, recurring infrastructure utilisation through OrbitLab and OrbitVault**.

3. Missions & Technology Validation

Mission	Outcome	Investment interpretation
RSEM — Jan 2024	Radiation-shielding experiment	Evidence of technology development / space qualification
MOI-TD — Dec 2024	**Partial success** — >20 experiments conducted, but the two primary objectives of imaging and on-orbit AI inference were not achieved	Important technical setback; software pipeline failure prevented the AI hardware from being triggered
MOI-1 — Jan 2026	**Failed/lost due to launch-vehicle anomaly**	Demonstrates launch/replacement-capital risk
MOI-1A — Oct 2026	Replacement mission	**Critical validation milestone**: whether the company can actually demonstrate the commercial orbital-AI proposition

The MOI-1 failure is important for an investment committee because it separates technology risk from launch risk. The company's own mission page states that MOI-1 recorded no fault through the failure event. The replacement mission creates a second opportunity to validate the product, but it also delays commercial scale and consumes capital.

MOI-1A: Transition from Technology Demonstration to Commercial Validation

On **2 October 2026 (IST)**, TakeMe2Space's **MOI-1A (My Orbital Infrastructure-1A)** was successfully launched aboard SpaceX's Transporter-18 mission and deployed into an orbital transfer vehicle. The spacecraft is designed to carry **117 TOPS of onboard computing, 16 GB RAM and a 9-band multispectral imager**, enabling Earth-observation data to be processed using AI directly in orbit rather than transmitting large volumes of raw data to Earth. Unlike the earlier MOI-TD mission, whose own technical documentation states that its primary imaging and on-orbit AI-inference objectives were not achieved because of a software pipeline issue, MOI-1A represents the company's next attempt to validate the technology in an operational environment. **However, launch success should not yet be interpreted as technology or commercial validation**: MOI-1A still needs to complete orbital injection, commissioning and end-to-end AI-processing tests. TakeMe2Space has reported **23 customers** across sectors including agriculture, mining, supply-chain intelligence and insurance for the mission, but public information does not yet disclose their contract values or committed utilisation. Accordingly, MOI-1A should be viewed as a **critical validation milestone**: successful in-orbit AI inference followed by repeatable customer utilisation would materially strengthen the case for TakeMe2Space's proposed orbital-compute-as-a-service model, while failure to demonstrate reliable compute, imaging or meaningful customer utilisation would remain a key thesis-breaker.

Planned Mission Roadmap

TakeMe2Space's disclosed roadmap indicates a progressive scale-up from smaller technology-demonstration spacecraft toward higher-capacity orbital infrastructure. **MOI-1B**, planned for Q4 2026, is a 30 kg platform with 20 W nominal and 50 W peak power, while **AADUSat, MOI-4 and MOI-5** are planned as approximately 100 kg platforms with 200 W nominal and 700 W peak power. The roadmap currently targets **AADUSat for Q1 2027 and MOI-4/MOI-5 for Q3 2028**. The increase in spacecraft mass, available payload capacity and power is important because sustained onboard AI processing and more capable sensing payloads require substantially greater electrical and thermal capacity. This roadmap therefore represents TakeMe2Space's intended transition from individual technology demonstrations toward a **scalable orbital-computing infrastructure network**.

4. Funding History & Use of Capital

Round	Amount	Lead / investors	Stated purpose / use
Pre-seed | Mar 2025	₹5.5Cr (~$0.63M)	SeaFund; Blume Ventures; Artha Venture Fund; AC Ventures; angels	Launch MOI-1, scale operations, expand regions/product portfolio, strengthen R&D and customer acquisition.
Seed | Jan 2026	$5M	Chiratae Ventures; Unicorn India Ventures; Artha Venture Fund; SeaFund	Move from single-satellite solution toward a networked satellite solution / constellation.

The capital trajectory is consistent with a hardware DeepTech company moving from technical validation toward infrastructure deployment. EntrepreneurLoop article attributed an indicative allocation of roughly 40% of funding to hardware, 30% to ground infrastructure, 20% to operations and 10% to regulatory/compliance.

5. Financial Performance

Publicly available financial information is limited. Inc42 reports FY25 revenue of ₹21.3 lakh. More detailed audited financial statements, gross margin, EBITDA, cash burn, receivables, working-capital requirements and customer-level revenue are not publicly available in the material reviewed.

Metric	Publicly available figure	Evidence quality
FY25 revenue	₹21.3 lakh	Third-party company database; period-specific.
Revenue-to-date	₹3.18Cr	Company-reported; announced Jan 2026; cumulative period not reconciled to FY25.
Revenue pipeline	₹20Cr	Company-reported; pipeline is not booked revenue.
Customers	23	Company / Reuters-reported; customer revenue and ARR not disclosed.
Total disclosed funding	~$5.6M+	Derived from ₹5.5Cr pre-seed plus $5M seed; FX and timing create small variation.
Profitability	Not publicly established	No reliable public P&L series sufficient for current profitability assessment.

The low disclosed historical revenue relative to funding is not automatically a negative signal for a DeepTech company because capital is being used to build and qualify physical infrastructure. The relevant financial question is whether the assets can generate sufficiently high utilisation and contribution margin once deployed.

6.Competitive Advantage, Cost Advantage & Potential Moats:

1. Orbital compute instead of raw-data downlink

- Traditional Earth-observation architecture follows **Capture → Store → Downlink → Ground Processing → Analysis**.
- TM2S changes this to **Capture → On-board AI Processing → Filter → Downlink only actionable intelligence**.
- This addresses the **downlink bottleneck**: satellites can generate more data than they can economically transmit to Earth because downlink windows, ground-station availability and bandwidth are constrained.
- TM2S states that its architecture can process **terabytes of data in orbit while transmitting only the resulting insights**, with claimed transmission-cost reductions of **60–94%** on its technology page and **up to 85%** on its OrbitLab page. These should be treated as **company claims rather than independently verified savings**.
- The economic advantage is potentially significant because customers do not pay to transmit and process large volumes of irrelevant pixels when the required output may be a small structured result such as "3 ships detected" or "crop stress identified."
- **Potential moat:** integration of sensing, onboard compute, AI inference and selective downlink into one operational workflow rather than simply providing imagery.

2. Latency Advantage

- **Processing at source:** AI runs onboard the satellite, eliminating the need to downlink large raw datasets before analysis.
- **Lower latency:** TM2S claims **5–15 minutes** from image capture to processed insight, versus **24–72 hours** for conventional workflows.
- **Faster decisions:** Lower latency can improve time-sensitive applications such as **disaster response, crop monitoring, maritime surveillance and infrastructure monitoring**, while simultaneously reducing downlink requirements.

3. Satellite-Time-as-a-Service / shared infrastructure

- The conventional model requires an organisation to fund **satellite CapEx, payload development, launch, ground infrastructure, operations and maintenance** before receiving useful data.
- TM2S instead attempts to sell **orbital capacity as a service**, allowing multiple customers to share the same satellite infrastructure.
- Chiratae explicitly compares this model with the transition from **on-premise servers to cloud computing**: customers pay for satellite utilisation rather than owning the underlying infrastructure.
- OrbitLab currently advertises **$4/minute**, or **$375 for a full orbit**, with academic access at **$200/orbit**. Enterprise customers can obtain volume pricing or dedicated capacity.
- This changes the customer's economics from **CapEx → OpEx**, potentially expanding the addressable market to organisations that cannot justify owning a satellite.
- **Potential moat:** once customers integrate TM2S's API/workflows into operational processes, switching may involve model re-validation, mission planning, data pipelines and regulatory/technical integration. However, this switching-cost moat is **not yet proven at scale**.

4. Constellation economics

- A **constellation** is a coordinated network of multiple satellites designed to provide greater geographic coverage, revisit frequency and service availability than a single satellite.
- TM2S plans a **six-satellite constellation**, with the company targeting daily global coverage and approximately **5 kW of aggregate onboard compute** according to reporting around its $5 million funding round.
- The economic logic is important: instead of one customer bearing the full fixed cost of a satellite, **multiple customers share the fixed infrastructure**.

- As the constellation expands, the company can potentially increase - geographic coverage, satellite availability, number of customer tasks, total billable compute minutes, data collected and processed, redundancy across spacecraft.
- This creates potential **operating leverage**: satellite manufacturing and launch costs are largely fixed for each spacecraft, while incremental customer utilisation can generate additional revenue without proportionally increasing the underlying infrastructure cost.
- However, the key metric to validate this thesis is **utilisation per satellite**. A large constellation with low utilisation would destroy the expected operating leverage.

5. Vertical integration reduces cost and iteration time

- TM2S states that it designs and builds the **satellite bus, subsystems, AI-Cube and imager in-house**, rather than relying extensively on third-party suppliers.
- Its catalogue spans **36 products**, including satellite platforms, onboard computers, AI processing, ADCS, power, communications, imaging and radiation protection.
- Chiratae highlights that the team has already developed **15+ satellite modules in-house**.
- Vertical integration can provide three economic benefits:
  1. **Lower procurement mark-ups** by removing intermediary supplier margins;
  2. **Shorter lead times** because engineering decisions do not depend on multiple external suppliers;
  3. **Faster iteration** because hardware and software teams can optimise the entire spacecraft together.
- TM2S itself argues that controlling the stack eliminates supplier delays and allows faster build cycles.
- **Potential moat:** proprietary know-how accumulates across multiple satellite generations, making the company's internal engineering system increasingly difficult to replicate. The moat becomes stronger if this integration consistently produces **lower cost per satellite, faster time-to-orbit and higher mission reliability**.

6. Radiation shielding as a hardware-level advantage

- Space electronics face **Total Ionizing Dose (TID)** and other radiation effects that can degrade conventional terrestrial electronics.
- TM2S's **RadShield** is a proprietary high-Z material coating designed to reduce radiation exposure to electronics.
- The company claims **up to 10× TID reduction** and states that components rated for months can potentially achieve approximately **five-year mission life** under its protection.
- If independently validated, this could reduce the need for expensive radiation-hardened components and enable the use of commercially available processors such as the **NVIDIA Jetson Orin NX** in LEO.
- This could create a cost/performance advantage because commercial processors generally offer much higher computing capability per dollar and watt than traditional space-qualified computing hardware.
- **Potential moat:** materials know-how + flight heritage + qualification data. The critical question is whether RadShield's performance remains differentiated after competitors can replicate the underlying approach.

7. Compute-per-watt and onboard AI

- Orbital computing is constrained by **power, thermal dissipation, radiation and mass**.
- TM2S's MOI platform combines a **117-TOPS NVIDIA Jetson Orin NX**, **16 GB LPDDR5**, **2 TB onboard storage**, and a stated **10–55 W AI-Cube power envelope**.
- This is important because every watt consumed by computing competes with the satellite's other requirements such as communications, attitude control, thermal management and imaging.
- A more efficient compute architecture allows more AI processing within the same spacecraft power budget.
- The competitive advantage therefore isn't simply “having a GPU in space”; it is the ability to make **compute + power + thermal management + radiation protection + pointing + imaging + communications** work reliably as an integrated spacecraft system.

8. Precision pointing + sensing + compute integration

- TM2S's MOI platform combines a **9-band multispectral imager**, precision pointing and onboard AI.
- The company states **<0.025° pointing accuracy (1σ)** and a **3°/second slew rate**, supported by a star tracker and four reaction wheels.
- This matters because orbital AI is only valuable if the satellite can reliably **point at the correct Area of Interest → capture usable imagery → process it → identify relevant information → transmit the result**.
- Integration of these functions can create a systems-engineering advantage over companies that specialise in only one layer of the stack.
- Again, this becomes a genuine moat only if repeated missions demonstrate superior reliability and cost.

9. Flight heritage and accumulated engineering data

- In DeepTech, **flight heritage** can itself become an important competitive asset.
- Components that have actually operated in LEO generate data on radiation exposure, thermal behaviour, vibration, power consumption, software reliability and subsystem interactions.
- TM2S has conducted multiple missions and says its MOI-TD completed **394 orbits**, while the company describes the mission as having validated its onboard compute and supporting systems. However, its own detailed technical documentation has also disclosed that the primary imaging and AI-inference objectives were not achieved, so this should be presented as **partial technology validation rather than an unqualified proof point**.
- Each additional successful mission can therefore reduce technical uncertainty for the next spacecraft and potentially shorten qualification cycles.
- **Potential moat:** accumulated proprietary engineering and environmental data that competitors cannot obtain without putting comparable hardware into orbit themselves.

10. Launch-cost advantage through rideshare and shared missions

- **Launch cost** is one of the major fixed costs of a satellite mission. Instead of buying an entire rocket, small satellites can use **rideshare**, sharing launch capacity with other spacecraft.
- TM2S's small-satellite architecture is compatible with this model and its hosted-payload offering allows multiple payloads to share spacecraft infrastructure.
- This can lower the **launch cost per customer** because the cost of reaching orbit is distributed across multiple payloads/customers.
- Hosted payloads create a second layer of utilisation: the company can monetise available **mass, volume and power** on a satellite that would otherwise remain unused.
- The resulting economic flywheel is:
  More payloads → better spacecraft utilisation → more revenue per launch → lower effective infrastructure cost per customer.
- The limitation is that launch cost alone does not determine satellite economics; manufacturing, insurance, ground stations, mission operations, replacement satellites and regulatory compliance remain significant costs.

11. Customer and developer ecosystem

- OrbitLab is designed as a software-access layer rather than requiring customers to interact directly with complex spacecraft infrastructure.
- Customers can define an **Area of Interest (AOI)**, configure imaging parameters, upload models and receive processed outputs through a software interface.
- This abstraction is strategically important: the company is attempting to make orbital infrastructure behave more like **cloud infrastructure/API access** than traditional aerospace procurement.
- If developers build applications specifically for TM2S's environment, the resulting ecosystem could create a **platform moat** through integrations, reusable models, customer workflows and accumulated operational knowledge.

Key Diligence Points:

The moat should therefore be treated as **"potentially compounding," not yet proven**. The evidence that would convert the above advantages into a true competitive moat would be:

1. **Lower cost per satellite** versus comparable competitors;
2. **Higher compute-per-watt** in actual LEO operations;
3. **Successful repeat missions** demonstrating reliability;
4. **High satellite utilisation** and paid compute minutes;
5. **Lower customer cost per useful insight**, not merely lower transmission cost;
6. **Repeat customers and increasing workloads**;
7. **Proprietary flight/operational data** that improves subsequent missions;
8. **Increasing constellation coverage and availability**;
9. **Evidence that customers cannot easily substitute TM2S with terrestrial cloud + existing EO providers**; and
10. **Improving unit economics as the constellation scales**.

7. Regulation, Data Governance & Regulatory Sandbox

Orbital computing creates a regulatory intersection rather than a single regulatory category. A satellite can simultaneously be a space object, communications platform, computing environment and data repository. TakeMe2Space itself has argued that India's existing frameworks were not designed specifically for this combination.

Framework	Relevance to orbital compute
Indian Space Policy 2023	Core policy architecture for private participation in space activities.
IN-SPACe 2024 NGP	Authorisation and supervision framework for private space activities.
DPDP Act 2023 + DPDP Rules 2025	Potential relevance where personal data is processed or stored.
Information Technology Act 2000	Digital/legal framework relevant to computing and data activities.
CERT-In Directions 2022	Cybersecurity incident reporting and related compliance considerations.

The company's proposed solution is a regulatory sandbox. Instead of waiting for a comprehensive orbital-data-centre regime, a supervised sandbox could test in-orbit storage, AI inference, disaster recovery, data security and compliance controls. The company has proposed a Baseline Orbital Compute Assurance (BOCA) working standard as a possible foundation for future technical and compliance guidance.

For investors, regulation is both a risk and a potential moat. Early engagement with regulators could lower uncertainty for the entire category, while compliance architecture could become a differentiator for financial, defence and other regulated customers.

8. Indian Peer Benchmark

Indian Peer Benchmark

Company	Orbital Compute	Compute-as-a-Service	EO + Onboard AI	Relationship to TM2S
TakeMe2Space	✓	✓	✓	**Reference company**
Pixxel + Sarvam AI	✓	Not publicly established	✓	**Closest emerging technology competitor**
Agnikul + NeevCloud	Planned	Not publicly established	AI inference	**Emerging technology competitor**
Pixxel	✓	Mainly EO/data products	✓	**Indirect competitor / substitute**
GalaxEye	Onboard processing / EO	No	✓	**Indirect competitor / substitute**
EON Space Labs	No	No	✓	**Adjacent payload/sensing player**

The Indian market does not yet appear to have a peer offering the **same publicly disclosed usage-based orbital Compute-as-a-Service model** as TM2S. The closest Indian comparisons are currently technology-level competitors rather than fully comparable business models. This supports the view that the domestic category is **early and fragmented**, while also highlighting the risk that existing EO and space-infrastructure companies could expand into orbital compute.

9. International Peer Validation — Aethero

Since detailed financial information and a truly comparable Indian company with an identical **orbital Compute-as-a-Service (CaaS)** model are not publicly available, the analysis extends to international peers to assess the commercial viability of TM2S's underlying business model.

Aethero (US) is the closest international business-model comparable identified. Its model combines space-compute hardware, onboard AI, orbital infrastructure and **Compute-as-a-Service**, allowing customers to deploy software workloads onto computing infrastructure in orbit. This closely mirrors TM2S's OrbitLab proposition.

Metric	Aethero	TakeMe2Space	Implication
Business model	Orbital Compute + CaaS/PaaS	Orbital Compute + OrbitLab	**Highly comparable**
Processor	NVIDIA Jetson Orin NX	NVIDIA Jetson Orin NX	**Same core compute architecture**
Compute capacity	**157 TOPS**	**117 TOPS**	Aethero ~34% higher advertised capacity
Customer workloads	Containerised applications	AI models/workloads	**Similar software layer**
Usage-based compute	✓	✓	**Direct model overlap**

Metric	Aethero	TakeMe2Space	Implication
EO + onboard processing	✓	✓	Similar use case
CaaS pricing	**$750/hour minimum**	**$4/minute (~$240/hour)**	TM2S advertises lower pricing*
Reported revenue	**>$2M**	Not publicly disclosed on a comparable basis	**External commercial validation**
Profitability	**Not publicly disclosed**	Not publicly disclosed	**Cannot assess comparative profitability**
Team size	~9 full-time employees + contractors*	Not publicly disclosed in comparable terms	Specialised-team model
Flight validation	Phobos mission	MOI-1A	Aethero currently has earlier CaaS flight heritage

*Pricing and team-size figures are not directly comparable because service specifications, minimum commitments and organisational structures differ.

The technological similarity is notable. **TM2S's MOI-1A uses an NVIDIA Jetson Orin NX-based AI-Cube delivering 117 TOPS**, while Aethero's Phobos mission deployed an NVIDIA Orin NX-based computing module delivering up to **157 TOPS**. Aethero also deployed multiple customer workloads through its orbital CaaS/PaaS architecture, providing evidence that orbital computing can be offered as a service rather than only as satellite hardware.

Aethero's co-founder **Edward Ge publicly reported that the company crossed $2 million in revenue in less than a year after its first mission, excluding SBIR funding**. This is a founder-reported figure rather than independently audited financial information and should therefore be treated as an indicative commercial-traction datapoint. **Aethero's profitability, margins and detailed financial statements were not publicly available**, preventing a meaningful comparison of profitability or unit economics

Investment Relevance

Aethero provides **external validation of both the technology and business model** underlying TM2S. The existence of a comparable company with flight-proven orbital computing, customer workloads and reported commercial revenue suggests that orbital CaaS can evolve beyond a technology demonstration into a monetisable infrastructure business.

TM2S's advertised **$4/minute OrbitLab pricing** is below Aethero's published $750/hour minimum; however, this should not be interpreted as a definitive cost advantage because the two offerings may differ in compute capacity, satellite access, minimum commitments and included services. The key diligence question for TM2S is whether its vertical integration and Indian cost base can translate into **lower cost per useful compute minute, higher satellite utilisation and attractive customer unit economics**.

Overall, Aethero validates the commercial plausibility of the category, but not TM2S's profitability or future success. The absence of publicly available profitability data for Aethero also highlights an important diligence gap: TM2S will need to demonstrate that its orbital infrastructure can achieve sufficient utilisation and pricing to generate attractive margins as the constellation scales.

10. Potential customer segments

Segment	Potential use case	Why orbital inference could matter
Agriculture	Crop stress / yield / field monitoring	Faster alerts; potentially less raw data transfer
Mining	Site monitoring / asset change detection	Frequent monitoring and anomaly detection
Supply chain / logistics	Asset / container / route monitoring	Near-real-time event detection
Insurance	Property / disaster / risk imagery	Faster claims/risk intelligence
GIS / geospatial companies	Model execution on satellite data	Infrastructure access without owning satellite
Research / universities	Testing AI workloads in orbit	Lower barrier to space experimentation
Government / defence	Persistent monitoring / resilient infrastructure	Potential sovereign and resilience value; requires rigorous compliance and procurement diligence
Satellite operators	Deploy own models in orbit	Potential reduction in downlink and processing burden

11. Illustrative Financial Model

1. Revenue & Cost Assumptions

Revenue Assumptions

Assumption	Base Case	Basis / Derivation
Customers on MOI-1A	23	Company-reported; Reuters also reported 23 customers
Current OrbitLab price	$4/minute	TakeMe2Space website
Full-orbit price	$375/orbit	TakeMe2Space website
Reported revenue pipeline	₹20 Cr	Company disclosed
Revenue till date	₹3.18 Cr	Company disclosed
Planned constellation	6 satellites	Company-stated plan

Customer Usage Assumption

Average revenue per customer is not publicly disclosed. Therefore, revenue is derived bottom-up using OrbitLab's published pricing and an assumed level of customer utilisation.

Base case:

500 billable minutes/customer/month × $4/minute × 12 months × ₹85/USD

= ₹20.4 lakh annual revenue/customer

The model assumes 10 of the 23 reported customers become meaningful paying/recurring customers in the base case. This is an analyst assumption rather than a company-reported figure.

Revenue Model

Revenue Driver	Calculation	Base Case Revenue
OrbitLab	10 customers × 500 min/month × $4 × ₹85	₹2.04 Cr
Pipeline conversion	₹20 Cr × 30%	₹6.00 Cr
Other businesses	Satellite/subsystem sales + hosted payloads + other services	₹2.00 Cr
Total Revenue		₹10.04 Cr

The ₹20 crore pipeline is treated as a commercial opportunity rather than booked revenue. A 30% conversion rate is an analyst assumption used to avoid treating the entire pipeline as realised revenue.

2. Cost Assumptions

Detailed satellite-level operating costs and profitability are not publicly disclosed. Therefore, costs are estimated using a simple employee-cost and operating-expense approach.

TakeMe2Space's current publicly disclosed team size is approximately 34 employees.

Employee Cost

Assumed blended employee cost = ₹15 lakh/employee/year

34 employees × ₹15 lakh

= ₹5.1 Cr annual employee cost

The ₹15 lakh blended cost is an analyst assumption based on the mix of technical and non-technical roles and external Indian aerospace compensation benchmarks.

Other Operating Costs

Ground operations, software/cloud infrastructure, administration, customer support and other recurring costs are assumed at ₹1.5 Cr annually in the base case.

Cost	Base Case	Basis
Employee cost	₹5.1 Cr	34 employees × ₹15L
Ground/operations/software/admin	₹1.5 Cr	Analyst assumption
Total operating cost	₹6.6 Cr	Derived

Satellite construction and launch CapEx is not included in EBITDA. It should be analysed separately as part of the capital required to build and scale the constellation.

3. Scenario Analysis

₹ Cr	Worst Case	Base Case	Best Case
Active paying customers	5	10	15
Billable minutes/customer/month	250	500	1,000
OrbitLab revenue	₹0.64	₹2.04	₹6.12
Pipeline conversion	10%	30%	60%
Pipeline revenue	₹2.0	₹6.0	₹12.0
Other revenue	₹1.0	₹2.0	₹4.0
Total Revenue	₹3.64 Cr	₹10.04 Cr	₹22.12 Cr
Operating costs	₹6.0	₹6.6	₹8.0
Illustrative EBITDA	-₹2.36 Cr	₹3.44 Cr	₹14.12 Cr

Worst Case

The company converts only a small portion of its reported pipeline and customer utilisation remains low. Only five customers become meaningful paying customers, with 250 billable minutes per customer per month. High operating costs relative to revenue result in an estimated EBITDA loss of approximately ₹2.4 crore.

Base Case

Ten customers become meaningful recurring users, with 500 billable minutes per customer per month. Approximately 30% of the reported ₹20 crore pipeline converts into revenue, supplemented by satellite/subsystem sales and hosted payloads. This results in estimated revenue of approximately ₹10 crore and illustrative EBITDA of approximately ₹3.4 crore.

Best Case

Fifteen customers become recurring users with substantially higher utilisation of 1,000 billable minutes per customer per month. Approximately 60% of the reported pipeline converts, while other businesses contribute additional revenue. This results in estimated revenue of approximately ₹22 crore and illustrative EBITDA of approximately ₹14 crore.

The best-case scenario assumes that TakeMe2Space successfully transitions from a technology-development company into a recurring orbital-infrastructure platform with increasing satellite utilisation.

4. Key Investment Metric — Satellite Utilisation

The most important operating variable in the model is monetised orbital utilisation.

Having six satellites does not automatically translate into six times the revenue. Customers must actually purchase and consume orbital capacity.

Therefore, the key metrics to monitor are:

- Billable minutes per satellite
- Revenue per satellite
- Revenue per customer
- Customer conversion from pipeline
- Repeat customer utilisation
- Cost per satellite deployed
- Operating cost per satellite

- Incremental revenue generated by additional orbital capacity

The investment thesis becomes stronger if customer utilisation increases faster than the cost of adding orbital capacity, creating operating leverage as the constellation scales.

5. Key Risks

1. Customer Conversion Risk

The company has reported 23 customers for MOI-1A, but public information does not establish how many are paying customers, their contract values or their expected recurring utilisation. Customer count is therefore not treated as equivalent to revenue.

2. Pipeline Conversion Risk

TakeMe2Space has disclosed a ₹20 crore revenue pipeline. The model does not treat this as booked revenue; instead, scenario-specific conversion rates are applied. Actual timing and conversion could differ materially.

3. Utilisation Risk

OrbitLab is a usage-based model. Revenue depends on customers actually consuming satellite capacity. Low utilisation could leave expensive orbital infrastructure underused.

4. Pricing Risk

The $4/minute and $375/orbit figures are current public list prices. Enterprise customers may receive volume pricing or use dedicated/custom deployments, meaning realised revenue per minute may differ.

5. Technology and Mission Risk

Successful launch does not by itself establish sustained commercial performance. Orbital commissioning, onboard AI processing, imaging, communications and customer workloads must operate reliably.

6. Capital Intensity

Scaling from individual satellites to a six-satellite constellation requires continued capital expenditure. Revenue growth must eventually outpace the capital required to add orbital capacity.

7. Cost Uncertainty

Detailed satellite-level operating costs, gross margins, customer acquisition costs and recurring operating expenses are not publicly disclosed. Cost assumptions in this model are therefore illustrative rather than company guidance.

6. Evidence Classification

Item	Classification
23 reported customers	Company-reported / externally reported
$4/minute OrbitLab price	Company-reported
$375/orbit price	Company-reported
₹20 Cr pipeline	Company-reported
₹3.18 Cr revenue till date	Company-reported
Six-satellite constellation	Company-stated plan
10 active paying customers	Analyst assumption
500 billable minutes/customer/month	Analyst assumption
30% pipeline conversion	Analyst assumption

Item	Classification
₹15L blended employee cost	Analyst assumption / external benchmark
₹1.5 Cr other operating costs	Analyst assumption
₹2 Cr other revenue	Analyst assumption
Worst/Base/Best case outcomes	Derived scenario analysis

7. Disclaimer

This is an illustrative VC scenario model, not a financial forecast or management guidance. Public disclosure is insufficient to construct a fully bottom-up financial model of TakeMe2Space.

Reported company figures—including the ₹20 crore pipeline, ₹3.18 crore revenue till date, 23 customers and six-satellite constellation plan—are distinguished from analyst assumptions. Revenue is derived primarily from disclosed OrbitLab pricing and assumed customer utilisation, while employee and operating costs are estimated using current team size and external industry compensation benchmarks.

The purpose of the model is not to predict TakeMe2Space's actual revenue or profitability. Instead, it tests the economic conditions required for the business model to work: customer conversion, recurring utilisation, pricing, operating-cost control and efficient deployment of orbital capacity.

The principal thesis-breakers are sustained low satellite utilisation, weak conversion of the reported pipeline, technology or mission failures, higher-than-expected constellation capital requirements and inability to achieve sufficient revenue per satellite. Conversely, successful technology validation, recurring customer workloads, increasing utilisation and declining unit costs as the constellation scales would strengthen the underlying investment thesis.

12. Investment Conclusion

TakeMe2Space presents an **investible early-stage opportunity** at the intersection of space infrastructure, AI and the emerging orbital-computing market.
Its potential differentiation lies in combining onboard AI, proprietary satellite subsystems, radiation protection and usage-based orbital compute through OrbitLab.
The reported ₹20 crore pipeline and growing customer base provide early commercial validation, while the six-satellite roadmap creates potential for recurring, utilisation-driven revenue.
The investment case remains dependent on successful technology validation, pipeline conversion and improving utilisation of deployed satellite capacity.
Overall, the risk-reward profile is attractive for a high-conviction early-stage investor willing to underwrite technology and execution risk in return for exposure to a potentially new infrastructure category.

Annexure – 1 - Space-Tech Terminologies

- **LEO (Low Earth Orbit):** An orbit around Earth at roughly 160–2,000 km altitude, commonly used for Earth observation and communications satellites.
- **Constellation:** A coordinated group of satellites operating together to provide broader coverage, higher availability, or faster revisit times.
- **Revisit Rate:** How frequently a satellite or constellation can observe the same location on Earth.
- **Downlink:** Transmission of data from a satellite to a ground station on Earth.
- **Uplink:** Transmission of data or commands from a ground station to a satellite.
- **Ground Station:** A terrestrial facility equipped with antennas and systems to communicate with satellites.
- **On-Orbit / Edge Computing:** Processing data directly on a satellite rather than sending raw data to Earth for processing.
- **Compute-per-Watt:** The amount of computing performance a satellite can deliver for each watt of available power.
- **Payload:** The mission-specific equipment carried by a satellite, such as an imager, sensor, or AI processor.
- **Hosted Payload:** A customer's equipment or experiment carried aboard another operator's satellite.
- **Rideshare:** Launching multiple satellites on the same rocket to share launch costs.
- **Flight Heritage:** Evidence that a space technology or component has previously operated successfully in space.
- **TID (Total Ionizing Dose):** The cumulative radiation exposure that can degrade or damage electronic components in space.
- **ADCS (Attitude Determination & Control System):** The system that determines and controls a satellite's orientation in space.
- **AOI (Area of Interest):** A specific geographic region selected for satellite observation or analysis.
- **Satellite Utilisation:** The proportion of a satellite's available capacity or operating time that generates revenue.
- **Payload Capacity:** The mass, volume, power, and other resources available for mission-specific equipment.
- **Onboard AI Inference:** Running an AI model on satellite-generated data in orbit to produce insights before downlinking.
- **Radiation Shielding:** Materials or techniques used to protect spacecraft electronics from harmful space radiation.
- **Ground Resolution:** The smallest ground-level detail that a satellite imaging system can distinguish, usually measured in metres per pixel.
- **Swath:** The width of the Earth's surface captured by a satellite sensor during one pass.
- **Satellite Bus:** The core spacecraft platform providing power, communications, computing, thermal control, and other support systems.
- **CapEx-to-OpEx:** A business-model shift from customers purchasing expensive satellite infrastructure to paying for satellite capacity as an operating expense.
- **Latency:** The time between capturing data and delivering the processed result or actionable insight to the customer.

Appendix 2 - Sources

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